



SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING



KALAHSTHI RAMYA
192311149RA

QUESTIONS

20 / 20 completed

Overriding

OVERRIDING

Write a Java program to demonstrate method overriding with a simple calculator. Create a parent class called Calculator with methods such as add(), subtract(), multiply(), and divide(). Then, create a child class called ScientificCalculator that overrides the multiply() method to perform a more complex calculation.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         Calculator calc = new ScientificCalculator(); // Upcasting
8
9         System.out.print("Enter first number: ");
10        double a = sc.nextDouble();
11
12        System.out.print("Enter second number: ");
13        double b = sc.nextDouble();
14
15        System.out.println("\n=== Results ===");
16        System.out.println("Addition: " + calc.add(a, b));
17        System.out.println("Subtraction: " + calc.subtract(a, b));
18        System.out.println("Multiplication (Overridden): " + calc.multiply(a, b));
19        System.out.println("Division: " + calc.divide(a, b));
20
21        sc.close();
```

OUTPUT

```
Exception in thread "main" java.util.NoSuchElementException
    at
    java.base/java.util.Scanner.throwFor(Scanner.java:937)
    at java.base/java.util.Scanner.next(Scanner.java:1594)
    at
    java.base/java.util.Scanner.nextDouble(Scanner.java:2564)
    at Main.main(Main.java:10)
```

INPUT

10
5

